

Tutorial Quiz 3 Rubric

Q1] If $w = x^2 + y^2 + z^2$ and $z = x^2 + y^2$

find $\left(\frac{\partial w}{\partial z}\right)_x$

Here x & y are independent variables
with z dependent on x, y .

For us x is fixed.

Also, $z = x^2 + y^2 \Rightarrow y^2 = z - x^2$

Substituting in 'w' we get

$$\begin{aligned} w &= \cancel{x^2} + (z - \cancel{x^2}) + z^2 \\ &= z^2 + z \end{aligned}$$

Now $\left(\frac{\partial w}{\partial z}\right)_x = 1 + 2z \quad \text{————— (1)}$

Q2] If $w = x^2 + y - z + \sin t$ and $x + y = t$

find $\left(\frac{\partial w}{\partial z}\right)_{y,t}$

Here, x & z are variables
 y & t are fixed

t is dependent ; $t = x + y$

we need to get rid of 'x'

$$\text{from } t = x + y \Rightarrow x = t - y$$

Substituting in 'w':

$$w = (t - y)^2 + y - z + \sin t$$

but since y & t are fixed.

under derivation w.r.t. 'z'

the terms $(t - y)^2$, y , $\sin(t)$
all go to zero.

$$\therefore \text{Now } \left(\frac{\partial w}{\partial z} \right)_{y,t} = -1 \quad \text{————— (1)}$$

Q₃] If $z = x + f(u)$ where $u = x \cdot y$

show that $x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = x$

$$u = x \cdot y$$

$$\frac{\partial u}{\partial x} = y$$

$$\frac{\partial u}{\partial y} = x$$

$$z = x + f(u)$$

$$\frac{\partial z}{\partial x} = 1 + f'(u) \frac{\partial u}{\partial x}$$

$$\frac{\partial z}{\partial y} = f'(u) \frac{\partial u}{\partial y}$$

$$= 1 + f'(u) y$$

$$= f'(u) x$$

$$\begin{aligned} \text{Now, } x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} &= x(1 + f'(u)y) - y(f'(u)x) \\ &= x + \cancel{xy f'(u)} - \cancel{xy f'(u)} = x \end{aligned}$$

————— (1)